

Homework_5

October 16, 2025

0.1 AMS 518. Homework 5

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0.1.2 Problem 1

Suppose we know that the error in the risk quadrangle is the mean squared error, defined by

$$\mathcal{E}(X) = \lambda \|X\|_2^2 = E[X^2]$$

Use the relations in Diagram 3 to derive all remaining elements in the risk quadrangle (Regret, Statistic, Deviation, Error). Follow these steps:

1. Derive the **regret** in the risk quadrangle with the relation

$$\mathcal{V}(X) = \mathcal{E}(X) + E[X]$$

2. Derive the **risk** in the risk quadrangle with the relation

$$\mathcal{R}(X) = \min_C \{C + \mathcal{V}(X - C)\}$$

3. Derive the **statistic** in the risk quadrangle with the relations

$$S(X) = \arg \min_C \mathcal{E}(X - C), \quad S(X) = \arg \min_C \{C + \mathcal{V}(X - C)\}$$

4. Derive the **deviation** in the risk quadrangle with the relation

$$\mathcal{D}(X) = \mathcal{R}(X) - E[X]$$

0.1.3 Solution

We are given the error measure $\mathcal{E}(X) = \lambda \|X\|_2^2 = E[X^2]$. We will now derive the other three components of the risk quadrangle.

1. Derivation of Regret $\mathcal{V}(X)$

We use the given relation $\mathcal{V}(X) = \mathcal{E}(X) + E[X]$.

Substituting the definition of $\mathcal{E}(X)$:

$$\mathcal{V}(X) = E[X^2] + E[X]$$

2. Derivation of Risk $\mathcal{R}(X)$

The risk is defined as $\mathcal{R}(X) = \min_C \{C + \mathcal{V}(X - C)\}$. First, let's express $\mathcal{V}(X - C)$ using the formula for regret we just found.

$$\mathcal{V}(X - C) = E[(X - C)^2] + E[X - C]$$

Expanding the terms:

$$E[(X - C)^2] = E[X^2 - 2XC + C^2] = E[X^2] - 2CE[X] + C^2$$

$$E[X - C] = E[X] - C$$

So,

$$\mathcal{V}(X - C) = (E[X^2] - 2CE[X] + C^2) + (E[X] - C)$$

Now, substituting this into the expression for $\mathcal{R}(X)$:

$$\mathcal{R}(X) = \min_C \{C + (E[X^2] - 2CE[X] + C^2) + (E[X] - C)\}$$

$$\mathcal{R}(X) = \min_C \{C^2 - 2CE[X] + E[X^2] + E[X]\}$$

To find the value of C that minimizes this expression, we take the derivative with respect to C and set it to zero.

Let $f(C) = C^2 - 2CE[X] + E[X^2] + E[X]$.

$$\begin{aligned} \frac{df}{dC} &= 2C - 2E[X] = 0 \\ \implies C &= E[X] \end{aligned}$$

Now we plug this optimal value of C back into the expression to find the minimum value, which is $\mathcal{R}(X)$:

$$\mathcal{R}(X) = (E[X])^2 - 2(E[X])E[X] + E[X^2] + E[X]$$

$$\mathcal{R}(X) = (E[X])^2 - 2(E[X])^2 + E[X^2] + E[X]$$

$$\mathcal{R}(X) = E[X^2] - (E[X])^2 + E[X]$$

Since the variance is defined as $\text{Var}(X) = E[X^2] - (E[X])^2$, the **risk** is:

$$\mathcal{R}(X) = \text{Var}(X) + E[X]$$

3. Derivation of Statistic $S(X)$

The statistic $S(X)$ is the value of C that achieves the minimum in the risk calculation. We can find it using either of the two given relations.

Method 1: Using $S(X) = \arg \min_C \{C + \mathcal{V}(X - C)\}$. From our work in Step 2, we found that the expression was minimized when $C = E[X]$, hence:

$$S(X) = E[X]$$

Method 2: Using $S(X) = \arg \min_C \mathcal{E}(X - C)$.

$$\mathcal{E}(X - C) = E[(X - C)^2]$$

Let's minimize this with respect to C . Let $g(C) = E[(X - C)^2] = E[X^2] - 2CE[X] + C^2$.

$$\begin{aligned} \frac{dg}{dC} &= -2E[X] + 2C = 0 \\ \implies C &= E[X] \end{aligned}$$

Both methods confirm that the optimal **statistic** is the expected value

$$S(X) = E[X]$$

4. Derivation of Deviation $\mathcal{D}(X)$

The deviation is defined by the relation $\mathcal{D}(X) = \mathcal{R}(X) - E[X]$.

Using our result for the risk $\mathcal{R}(X)$ from Step 2

$$\mathcal{D}(X) = (\text{Var}(X) + E[X]) - E[X]$$

This simplifies to:

$$\mathcal{D}(X) = \text{Var}(X)$$

The **deviation** is the variance of X .

0.1.4 Problem 2

1. Use the dataset in the case study

https://uryasev.ams.stonybrook.edu/index.php/research/testproblems/financial_engineering/on-implementation-of-cvar-regression/.

We start as usual by setting up our working directory and checking if we have the data in the data file provided in the professor's website

```

> setwd("C:/Users/JUANJO/Documents/GitHub/ams-518/homework-5/data")
> # Set path to Rdata file including the RData file's name with CS:
> load("C:/Users/JUANJO/Documents/GitHub/ams-518/homework-5/data/
problem_CVaR2_err_075.RData")
> names(problem.list)
[1] "problem_name"      "problem_statement" "matrix_s"

```

Ok so we can see that we already have the matrix of scenarios and the problem statement loaded in the data file

2. Conduct CVaR regression with confidence level 0.7 using the code in Problem 1 of the case study.

In this problem, we regress CVaR of the return distribution of the Fidelity Magellan Fund on the explanatory variables: Russell Value Index (RUJ), Russell 1000 Value Index (RLV), Russell 2000 Growth Index (RUO) and Russell 1000 Growth Index (RLG).

```

> problem.list$problem_name
[1] "problem_1"
> # Check what value is stored for alpha
> problem.list$problem_statement
[1] "minimize" "cvar2_err(0.75,matrix_s)"
> # Replace alpha value
> problem.list$problem_statement[2] <- "cvar2_err(0.7,matrix_s)"
> # Check again to see if correct value was stored properly
> problem.list$problem_statement
[1] "minimize" "cvar2_err(0.7,matrix_s)"

```

Now we run the solver and store the results

```

> # Run PSG Solver to optimize problem stored in problem.list
> results_0.7 <- rpsg_solver(problem.list)
Data successfully read from C:\Aorda\PSG25\PSG25_license.bin
Running solver
Reading problem formulation
Asking for data information
Getting data
100% of matrix_s was read
Start optimization
Ext.iteration=0  Objective=0.141400532111E+00  Residual=0.000000000000E+00
Ext.iteration=65  Objective=0.116920944960E-01  Residual=0.000000000000E+00
Optimization is stopped
Solution is optimal
Calculating resulting outputs. Writing solution.
Objective: objective = 1.169209449596E-02 [2.959617030618E-10]
Solver has normally finished. Solution was saved.
Problem: problem_1, solution_status = optimal
Timing: data_loading_time = 0.05, preprocessing_time = 0.01, solving_time = 0.01
Variables: optimal_point = point_problem_1
Objective: objective = 1.169209449596E-02 [2.959617030618E-10]
Function: cvar2_err(0.7,matrix_s) = 1.169209449596E-02

```

```
Function: pseudo_R2_cvar2_err(0.7,matrix_s) = 8.576271843963E-01
Function: adjusted_pseudo_R2_cvar2_err(0.7,matrix_s) = 8.580780880086E-01
Function: contributions(cvar2_err(0.7,matrix_s)) = point_contributions_cvar2_err
OK. Solver Finished
```

As we can see, the values for the CVaR regression obtained are

```
> results_0.7$point_problem_1
      intercept      rlv      rlg      ruj      ruo
0.006215688  0.575909619  0.488580466 -0.073134971 -0.006187690
```

3. Conduct quantile regression with confidence levels 0.7, 0.75, 0.85, 0.9 and 0.95.

Now we proceed to run the following quantile regression, for this one we need to use the Koenker Bassett error function

```
# Quantile regression with confidence levels 0.7, 0.75, 0.85, 0.9 and 0.95
```

```
# alpha = 0.7
problem.list$problem_statement[2] <- "kb_err(0.7, matrix_s)"
results_quantile_0.7 <- rpsg_solver(problem.list)
```

```
# alpha = 0.75
problem.list$problem_statement[2] <- "kb_err(0.75, matrix_s)"
results_quantile_0.75 <- rpsg_solver(problem.list)
```

```
# alpha = 0.85
problem.list$problem_statement[2] <- "kb_err(0.85, matrix_s)"
results_quantile_0.85 <- rpsg_solver(problem.list)
```

```
# alpha = 0.9
problem.list$problem_statement[2] <- "kb_err(0.9, matrix_s)"
results_quantile_0.9 <- rpsg_solver(problem.list)
```

```
# alpha = 0.95
problem.list$problem_statement[2] <- "kb_err(0.95, matrix_s)"
results_quantile_0.95 <- rpsg_solver(problem.list)
```

After doing so we store the results and we can give a look onto those

```
q_0.7 <- results_quantile_0.7$point_problem_1
q_0.75 <- results_quantile_0.75$point_problem_1
q_0.85 <- results_quantile_0.85$point_problem_1
q_0.9 <- results_quantile_0.9$point_problem_1
q_0.95 <- results_quantile_0.95$point_problem_1

> q_0.7
      intercept      rlv      rlg      ruj      ruo
0.002059018  0.530025245  0.511601538 -0.062879035 -0.004608559
> q_0.75
      intercept      rlv      rlg      ruj      ruo
```

```

0.002953625 0.533445171 0.512397566 -0.062817452 -0.005050635
> q_0.85
  intercept          rlv          rlg          ruj          ruo
0.005106962 0.567370966 0.495101827 -0.073112157 -0.006047001
> q_0.9
  intercept          rlv          rlg          ruj          ruo
0.006705003 0.598016378 0.489920314 -0.098475905 -0.006678617
> q_0.95
  intercept          rlv          rlg          ruj          ruo
0.008982271 0.622364487 0.474911475 -0.098051076 -0.007805269

```

4. Take the average of the coefficients of the five quantile regressions, and compare it with the coefficient of CVaR regression.

Now we just average those and compare the results with the CVaR regression coefficients

```

> sum <- q_0.7 + q_0.75 + q_0.85 + q_0.9 + q_0.95
> average <- sum / 5
> average
  intercept          rlv          rlg          ruj          ruo
0.005161376 0.570244449 0.496786544 -0.079067125 -0.006038016

```

```
average <- average[-1]
```

```

cvar_0.7 <- results_0.7$point_problem_1
cvar_0.7 <- cvar_0.7[-1]

```

We can see that the relative error between these coefficients is

```

> relative_error = abs(cvar_0.7 - average) / abs(cvar_0.7)
> relative_error
          rlv          rlg          ruj          ruo
0.009836908 0.016795756 0.081112410 0.024188944

```

Hence, as we can see, the error is low, which means that the two results are close to each other, in this case we are getting a difference because the discretization we are using, in which we are taking 5 points to approximate the $CVaR_{0.7}$. The more points we take to run the quantile regression, the better the accuracy is going to be.